Supporting information for:
Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

This supporting information accompanies the manuscript titled *Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling* and provides expanded workflow notes and supplemental derivations.

S1 Staged refinement workflow (expanded)

1. **Beam characterization (direct beam).** Acquire direct-beam images at several detector distances. Fit the spot position/shape versus distance to determine the incident-beam widths (w_x, w_z) and Gaussian divergences $(\Delta\theta_x, \Delta\theta_z)$ that set the instrument-limited broadening.

2. **Detector geometry calibration (3D powder standard).** Measure a polycrystalline calibrant (e.g., hBN). Fit multiple Debye–Scherrer rings in detector coordinates and refine detector tilts (β, γ), in-plane detector rotation χ, and beam center (x_0, y_0) so the rings forward-project to concentric circles (equivalently, ellipses in the raw detector frame are corrected to circles).

3. **Sample alignment and goniometer misalignment (2D powder film).** Use the 2D-powder signature (paired symmetric in-plane features and single specular features) to seed alignment. With detector parameters fixed, refine sample alignment by minimizing positional residuals between measured and simulated features spanning specular and off-specular regions. Refine $(\theta_i, \delta, z_B, z_S)$ and the goniometer-axis misalignment rotations (α, ψ); include film dimensions (W, H) and thickness t to weight footprint and absorption.

4. **Mosaicity from line shapes (two θ_i images).** Acquire two images at different θ_i and fit them simultaneously using a shared out-of-plane misorientation kernel

$$\omega(\Delta\theta) = \eta\,\frac{1}{\pi}\frac{\Gamma}{\Delta\theta^2 + \Gamma^2} + (1-\eta)\,\frac{1}{\sqrt{2\pi}\sigma}\exp\!\left(-\frac{\Delta\theta^2}{2\sigma^2}\right), \qquad 0 \le \eta \le 1,$$

   (use the 2π-periodic wrapped form on $(-\pi, \pi]$; see Sec. S1.1). Determine parameters from arc/cap widths and specular off-condition tails after normalizing an overall scale. Report Lorentzian FWHM $= 2\Gamma$, Gaussian FWHM $= 2\sqrt{2\ln 2}\,\sigma$, and mixing factor η.

5. **Structure-factor constrained intensities (two θ_i images).** With geometry and mosaicity fixed, refine the intensity model against the full 2D images (including specular).

Evaluate all allowed reciprocal vectors G explicitly and use $|F_G|^2$ to set reflection strengths. Refine an overall scale plus structure-factor parameters (occupancies, anisotropic Debye–Waller terms, and allowed static displacements) to match reflection-to-reflection intensity ratios across both images.

6. **Stacking and polytype disorder (diffuse along Q_z).** After the ordered model converges, introduce stacking-disorder (layer-slip / polytype) parameters and refine them against the diffuse intensity between Bragg peaks along Q_z (diffuse intensity along specular / Bragg rods and its Q_z-dependence), e.g., with an Hendricks–Teller-type model when appropriate.

S1.1 Wrapped mosaic distribution

In the main text we describe out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$ in the crystallite tilt angle $\Delta\theta$ relative to the substrate normal. Since $\Delta\theta$ is an orientation angle, $\Delta\theta$ and $\Delta\theta + 2\pi$ represent the same physical state; accordingly, ω is most naturally treated as a 2π-periodic ("wrapped") probability density on $\Delta\theta \in (-\pi, \pi]$. Without wrapping, an otherwise smooth kernel on the real line produces an artificial seam at $\pm\pi$ and assigns different weights to physically equivalent orientations separated by 2π, which can bias the weak tail intensity used to constrain mosaicity.

We use a wrapped pseudo-Voigt kernel,

$$\omega_{\mathrm{w}}(\Delta\theta; \eta, \Gamma, \sigma) = \eta\, L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1-\eta)\, G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \le \eta \le 1,$$

where the wrapped Lorentzian and wrapped Gaussian may be written by periodic summation (equivalently, in closed form) as

$$L_{\mathrm{w}}(\Delta\theta; \Gamma) = \sum_{m=-\infty}^{\infty} \frac{1}{\pi} \frac{\Gamma}{(\Delta\theta + 2\pi m)^2 + \Gamma^2} = \frac{1}{2\pi} \frac{\sinh \Gamma}{\cosh \Gamma - \cos \Delta\theta},$$

$$G_{\mathrm{w}}(\Delta\theta; \sigma) = \sum_{m=-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta\theta + 2\pi m)^2}{2\sigma^2}\right] = \frac{1}{2\pi}\left[1 + 2\sum_{n=1}^{\infty} e^{-n^2\sigma^2/2} \cos(n\Delta\theta)\right] = \frac{1}{2\pi}\, \theta_3\!\left(\frac{\Delta\theta}{2}, e^{-\sigma^2/2}\right).$$

Both integrate to unity over $(-\pi, \pi]$, so ω_{w} is normalized. In the narrow-width regime relevant here ($\Gamma, \sigma \ll \pi$), ω_{w} reduces locally near $\Delta\theta = 0$ to the familiar unwrapped Lorentzian/Gaussian used in the main text, and we report Lorentzian FWHM $= 2\Gamma$ and Gaussian FWHM $= 2\sqrt{2\ln 2}\,\sigma$.

For a reflection of Bragg-sphere radius R, mosaicity redistributes intensity in polar angle ϑ with meridional line density $I_0\, \omega(\vartheta)\, d\vartheta$. Random in-plane rotations about the film normal revolve this line density about the G_z axis, yielding an axially symmetric surface density $\sigma(\vartheta)$. With area element $dA = 2\pi R^2 \sin\vartheta\, d\vartheta$, conservation of intensity gives

$$\sigma(\vartheta) = \frac{I_0}{2\pi R^2 \sin\vartheta}\, \omega(\vartheta), \qquad 0 < \vartheta < \pi,$$

as used in the main text.

S1.2 Wave propagation, refraction, and absorption

Wave attenuation occurs both at the interfaces (Fresnel transmission) and as the wave propagates through the film (absorption). In the forward model we therefore (i) refract each ray into and out of the film, (ii) assign the absorption to the out-of-plane component of the internal wavevector, and (iii) weight scattering from a finite thickness t by thickness-averaged Beer–Lambert attenuation. This subsection records the explicit equations used.

Complex refractive index. We write the complex refractive index as

$$\tilde{n} = 1 - \delta_n + i\beta_n, \tag{S1}$$

where the subscript n avoids confusion with the geometric angles (δ, β) used elsewhere. In terms of the wavelength λ, electron density ρ_e, classical electron radius r_e, and linear absorption coefficient μ,

$$\delta_n = \frac{r_e \lambda^2 \rho_e}{2\pi}, \qquad \beta_n = \frac{\mu\lambda}{4\pi}. \tag{S2}$$

Angles and refraction. Let $k_0 = 2\pi/\lambda$ and take $\hat{z}$ along the sample normal. For an incident ray in air, the grazing-incidence angle θ_i and in-plane azimuth φ_i may be expressed in terms of the external wavevector components as

$$\sin\theta_i = \frac{k_{i,z}}{k_0}, \tag{S3}$$

$$\tan\varphi_i = \frac{k_{i,x}}{k_{i,y}}. \tag{S4}$$

Upon refraction, we compute the transmitted grazing angle using the real part of the refractive index,

$$\theta_t = \cos^{-1}\left(\frac{\cos\theta_i}{\mathrm{Re}(\tilde{n})}\right). \tag{S5}$$

Internal wavevector components. We define a real internal wavevector magnitude for the in-plane components as

$$k' = k_0 \left(\mathrm{Re}\left(\tilde{n}^2\right)\right)^{1/2}, \tag{S6}$$

so that

$$k_{tx} = k' \cos\theta_t \sin\varphi_i, \tag{S7}$$

$$k_{ty} = k' \cos\theta_t \cos\varphi_i. \tag{S8}$$

To satisfy the boundary condition at the surface, we take only the out-of-plane component to be complex and write

$$k_{tz}^2 = a + ib, \qquad k_{tz} = u + iv = \sqrt{a + ib}. \tag{S9}$$

Expanding $(u + iv)^2$ gives

$$(u + iv)^2 = u^2 - v^2 + 2iuv = a + ib,$$

so that the real and imaginary parts satisfy

$$a = u^2 - v^2 = (u + v)(u - v), \qquad b = 2uv = (u + v)^2 - u^2 - v^2. \tag{S10}$$

The magnitude follows from

$$|k_{tz}|^2 = u^2 + v^2 = \sqrt{(a + ib)(a - ib)} = \sqrt{a^2 + b^2}, \tag{S11}$$

which yields

$$u^2 = \frac{\sqrt{a^2 + b^2} + a}{2}, \qquad v^2 = \frac{\sqrt{a^2 + b^2} - a}{2}. \tag{S12}$$

In our parameterization,

$$\mathrm{Re}(k_{tz}^2) = k'^2 \sin^2\theta_t \equiv a_t, \tag{S13}$$

$$\mathrm{Im}(k_{tz}^2) = k_0^2\ \mathrm{Im}\Big(\tilde{n}^2\Big) \equiv b, \tag{S14}$$

so that

$$\mathrm{Re}(k_{tz}) = \sqrt{\frac{\sqrt{a_t^2+b^2}+a_t}{2}}, \tag{S15}$$

$$\mathrm{Im}(k_{tz}) = \sqrt{\frac{\sqrt{a_t^2+b^2}-a_t}{2}}. \tag{S16}$$

We take the physical branch $\mathrm{Im}(k_{tz}) > 0$ so that the field decays into the film.

Exit path and refraction out of the film. For the scattered ray we repeat the same construction using the external grazing-exit angle and in-plane azimuth (θ_f, φ_f). We compute an internal exit angle θ_t' (again using $\mathrm{Re}(\tilde{n})$) and form

$$k_{tx}' = k' \cos\theta_t' \sin\varphi_f, \tag{S17}$$

$$k_{ty}' = k' \cos\theta_t' \cos\varphi_f, \tag{S18}$$

$$a_f \equiv k'^2 \sin^2\theta_t', \tag{S19}$$

with the same $b_f = b_t = b$,

$$\mathrm{Re}(k_{tz}') = \sqrt{\frac{\sqrt{a_f^2+b^2}+a_f}{2}}, \tag{S20}$$

$$\mathrm{Im}(k_{tz}') = \sqrt{\frac{\sqrt{a_f^2+b^2}-a_f}{2}}. \tag{S21}$$

Optical reversibility then gives the refraction back into air; inverting the relation for θ_t gives

$$\theta_f = \cos^{-1}\big(\mathrm{Re}(\tilde{n})\cos\theta_t'\big)\,. \tag{S22}$$

Fresnel transmission factors. We weight each ray by Fresnel transmission factors at entry and exit interfaces. For a scalar field written in terms of the normal wavevector components, one convenient form for an amplitude transmission coefficient from medium 1 to 2 is

$$t_{12} = \frac{2k_{1z}}{k_{1z}+k_{2z}}. \tag{S23}$$

Absorption weight (uniform scattering through thickness). Assuming a uniform probability of scattering across depth $z \in [0, t]$ and Beer–Lambert attenuation along the incident and exit paths, the intensity weight used in the forward model is

$$w_{\mathrm{abs}} = \frac{1}{t}\int_0^t \exp\left[-\mu z\left(\frac{1}{\sin\theta_i}+\frac{1}{\sin\theta_f}\right)\right] dz \tag{S24}$$

$$= \frac{1-\exp\left[-\mu t\left(\frac{1}{\sin\theta_i}+\frac{1}{\sin\theta_f}\right)\right]}{\mu t\left(\frac{1}{\sin\theta_i}+\frac{1}{\sin\theta_f}\right)}. \tag{S25}$$

S2 Material background and stacking–fault model

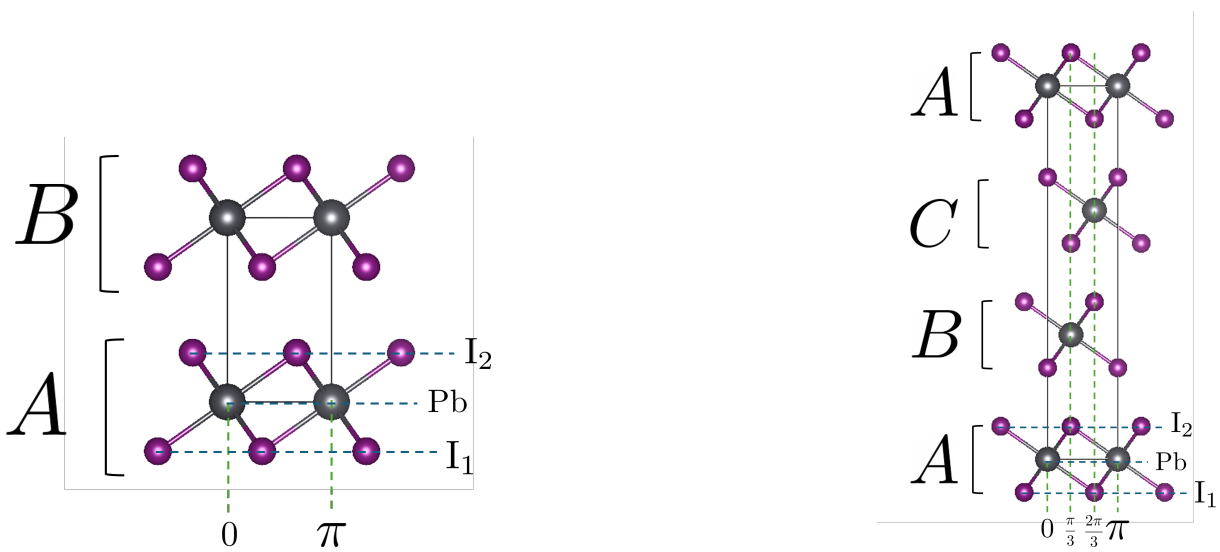
In layered metal–halides such as PbI_2 each I–Pb–I sandwich behaves like a rigid trilayer that can shift laterally with negligible energy cost compared with the much stronger intra-layer bonding. Consequently the crystal can adopt several polytypes that differ only in the stacking sequence of these sandwiches along the c axis. Here we focus on the two sequences that bracket the experimentally relevant window:

S2.1 Polytypism of PbI_2

2H ($P\bar{3}$m1) AB repeat stable below 140 °C; no lateral shift ($\phi_{AB} = 0$); see Fig. S1a.

6H ($R\bar{3}$) ABCA repeat above ≈ 200 °C; lateral shift $a/\sqrt{3}$ gives $\phi = 2\pi/3$; 2H ⇔ 6H transition near 94 °C; see Fig. S1b.

Our samples never achieve perfect periodicity; stacks often contain finite 2H and 6H "domains" interspersed by stacking faults. To model the resulting diffuse scattering we adapt the Hendricks–Teller (HT) Markov formalism and follow the streamlined matrix notation introduced a few years later by Lifshitz, who showed how to re-express pair probabilities through the eigenvalues of a 3×3 transition matrix.



(a) 2H polytype (AB) with no in-plane phase shifts.

(b) 6H R$\bar{3}$ polytype (ABCA) with $\frac{1}{3}$-lattice translations giving $e^{i2\pi/3}$ phase steps.

Figure S1: Low-temperature 2H and high-temperature 6H stacking sequences in PbI_2. Arrows illustrate the lateral shifts responsible for the phase factors used in the HT–Lifshitz model.

S2.2 Molecular structure factor for PbI_2

A single layer form factor is

$$F_m = b_{Pb} + b_I \left[e^{2\pi i(H/3+2K/3+Lz/3)} + e^{2\pi i(2H/3+K/3-Lz/3)} \right] \tag{S26}$$

Where now the amplitude will be

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \tag{S27}$$

Where $\vec{R_p} = m\vec{a} + n\vec{b}$ is the in-plane position vector, with $m, n \in \mathbb{Z}$. Then $\vec{Q}\cdot\vec{R_p} \to Hm + kn$, and the Bragg condition makes H, K integers.

S3 No-correlation stacking-fault model

Let $\tau_j \in \{0, 1\}$ indicate whether the j-th interface slips ($\tau_j = 1$) or not ($\tau_j = 0$). The structure factor of one layer is $F_0(h, k, \ell)$, and for a three-layer (ABC) repeat the per-layer phase advance is

$$\phi(h, k, \ell) = \frac{h + 2k}{3} + \ell/3, \tag{S28}$$

with h, k, ℓ in reciprocal-lattice units.
We now add the phase of each layer to the form factor from equation S27.

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \sum_{\ell} e^{i\phi(HKL)\ell} \tag{S29}$$

Writing $e^{i\phi\ell} = \prod_{j=1}^{\ell} e^{i\phi}$

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \prod_{\ell} e^{i\phi(HKL)\ell} \tag{S30}$$

S3.1 Coherent amplitude of an N-layer stack

At layer ℓ the factor multiplying F_0 is

$$\left[1 - \tau_\ell + \tau_\ell e^{-i\phi}\right] e^{i\phi\ell},$$

so that

$$A = F_0 \sum_{\ell=0}^{N-1} \left[\prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi})\right] e^{i\phi\ell}. \tag{S31}$$

S3.2 Uncorrelated faults

Assume the τ_j are i.i.d. with slip probability $p = \Pr(\tau_j = 1)$. Independence gives

$$\left\langle \prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi}) \right\rangle = \left[(1-p) + p\, e^{-i\phi}\right]^{\ell} \equiv f^{\,\ell}, \qquad f \equiv (1-p) + p\, e^{-i\phi}. \tag{S32}$$

S3.3 Mean amplitude

Averaging (S31) and inserting (S32) gives the geometric series

$$\langle A \rangle = F_0 \sum_{\ell=0}^{N-1} f^{\ell} e^{i\phi\ell} = F_0 \frac{1 - (f\, e^{i\phi})^N}{1 - f\, e^{i\phi}}. \tag{S33}$$

S3.4 Infinite-crystal limit

For any non-zero fault probability ($|f| < 1$),

$$\boxed{\lim_{N\to\infty} \langle A \rangle = F_0 \, \frac{f}{1 - f\, e^{i\phi}}} \tag{S34}$$

The perfect-crystal result is recovered as $f \to 1$.

S3.5 Mean-square amplitude

The heights of sharp Bragg peaks: i.e. the square of the coherently averaged structure amplitude. This is only the part of the intensity that survives perfect phase coherence: sharp Bragg peaks. Any layer-to-layer fluctuations are averaged before squaring and thus disappear.

$$\begin{aligned} |\langle A \rangle|^2 &= |F_0|^2 \frac{f^2}{|1 - f\, e^{i\phi}|^2} \\ &= |F_0|^2 \frac{f^2}{(1 - f\cos\phi)^2 + f^2 \sin^2\phi} && \text{(S35)} \\ &= |F_0|^2 \frac{f^2}{(1-f)^2 + 4f \sin^2 \dfrac{\phi}{2}}. && \text{(S36)} \end{aligned}$$

S3.6 Ensemble-averaged intensity

The full detector signal; adds the broadened peaks and diffuse background arising from stacking faults.

$$< |A|^2 > \tag{S37}$$

S3.6.1 Layer Autocorrelation

The layer-resolved autocorrelation of the stacking sequence tells you how phase coherence decays with layer phase shifts. It is the kernel that converts local disorder into the full intensity giving both Bragg and diffuse components.

Define

$$g_j \equiv 1 - \tau_j + \tau_j e^{-i\phi}, \qquad S_\ell \equiv \prod_{j=0}^{\ell} g_j. \tag{S38}$$

We wish to compute

$$\langle S_\ell S_{\ell'}^* \rangle = \langle \prod_{j=0}^{\ell} g_j \prod_{j'=0}^{\ell'} g_{j'}^* \rangle. \tag{S39}$$

Since the g_j are i.i.d., expectations of products factor, and $g_j g_j^* = 1$ identically. We split into two cases:

Case A: $\ell \geq \ell'$**.**

$$S_\ell S_{\ell'}^* \;=\; \underbrace{\Big(\prod_{j=0}^{\ell'} g_j\Big)\Big(\prod_{j'=0}^{\ell'} g_{j'}^*\Big)}_{=\,1\ \text{(common layers)}} \;\underbrace{\Big(\prod_{j=\ell'+1}^{\ell} g_j\Big)}_{\text{extra layers}} \tag{S40}$$

- For $0 \leq j \leq \ell'$, $g_j g_j^* = 1$, so the first two products give 1. - For $j = \ell'+1, \ldots, \ell$, independence implies

$$\Big\langle \prod_{j=\ell'+1}^{\ell} g_j \Big\rangle = \prod_{j=\ell'+1}^{\ell} \langle g_j \rangle = f^{\,\ell-\ell'}. \tag{S41}$$

Hence

$$\langle S_\ell S_{\ell'}^* \rangle = f^{\,\ell-\ell'}, \qquad (\ell \geq \ell'). \tag{S42}$$

Case B: $\ell < \ell'$**.** By an identical argument, swapping roles,

$$\langle S_\ell S_{\ell'}^* \rangle = \big(f^*\big)^{\ell'-\ell}, \qquad (\ell < \ell').$$

The Solutions are then

$$\boxed{\langle S_\ell S_{\ell'}^* \rangle = \begin{cases} f^{\,\ell-\ell'}, & \ell \geq \ell', \\ \big(f^*\big)^{\ell'-\ell}, & \ell < \ell'. \end{cases}} \tag{S43}$$

S3.7 Single-sum form of the double series

With $\ell'' = \ell - \ell'$, the double sum becomes

$$\sum_{\ell=0}^{N_z-1} \sum_{\ell'=0}^{N_z-1} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \sum_{\ell''=-N_z+1}^{N_z-1} f^{|\ell''|} e^{i(\phi-\psi)\ell''}. \tag{S44}$$

Splitting out $\ell'' = 0$ and the $\pm\ell''$ pairs yields

$$N_z \left[1 + \sum_{\ell''=1}^{N_z-1} f^{\ell''} \big(e^{i(\phi-\psi)\ell''} + e^{-i(\phi-\psi)\ell''} \big) \right],$$

with

$$\sum_{\ell''=1}^{N_z-1} f^{\ell''} e^{\pm i(\phi-\psi)\ell''} = \frac{f\, e^{\pm i(\phi-\psi)} \big[1 - f^{N_z-1} e^{\pm i(\phi-\psi)(N_z-1)} \big]}{1 - f\, e^{\pm i(\phi-\psi)}}.$$

S3.8 Infinite-crystal limit $(0 \leq f < 1)$

Letting $N_z \to \infty$ gives

$$\sum_{\ell,\ell'} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \, \frac{1 - f^2}{1 + f^2 - 2f\cos(\phi-\psi)}. \tag{S45}$$

S3.9 Scattering cross-section

Substituting (S45),

$$\boxed{\frac{d\sigma}{d\Omega} = |F_m|^2 \, \frac{(2\pi)^2}{A_c} \, \delta(\mathbf{Q} - \mathbf{G}_p) \, N_p N_z \, \frac{1 - f^2}{1 + f^2 - 2f\cos(\phi - \psi)}}. \tag{S46}$$

This is identical to the Hendrix–Teller formula upon identifying $\phi^{\mathrm{HT}} = \phi - \psi$ and noting that their parameter $\cos(\phi^1 - \phi^2)$ equals f.